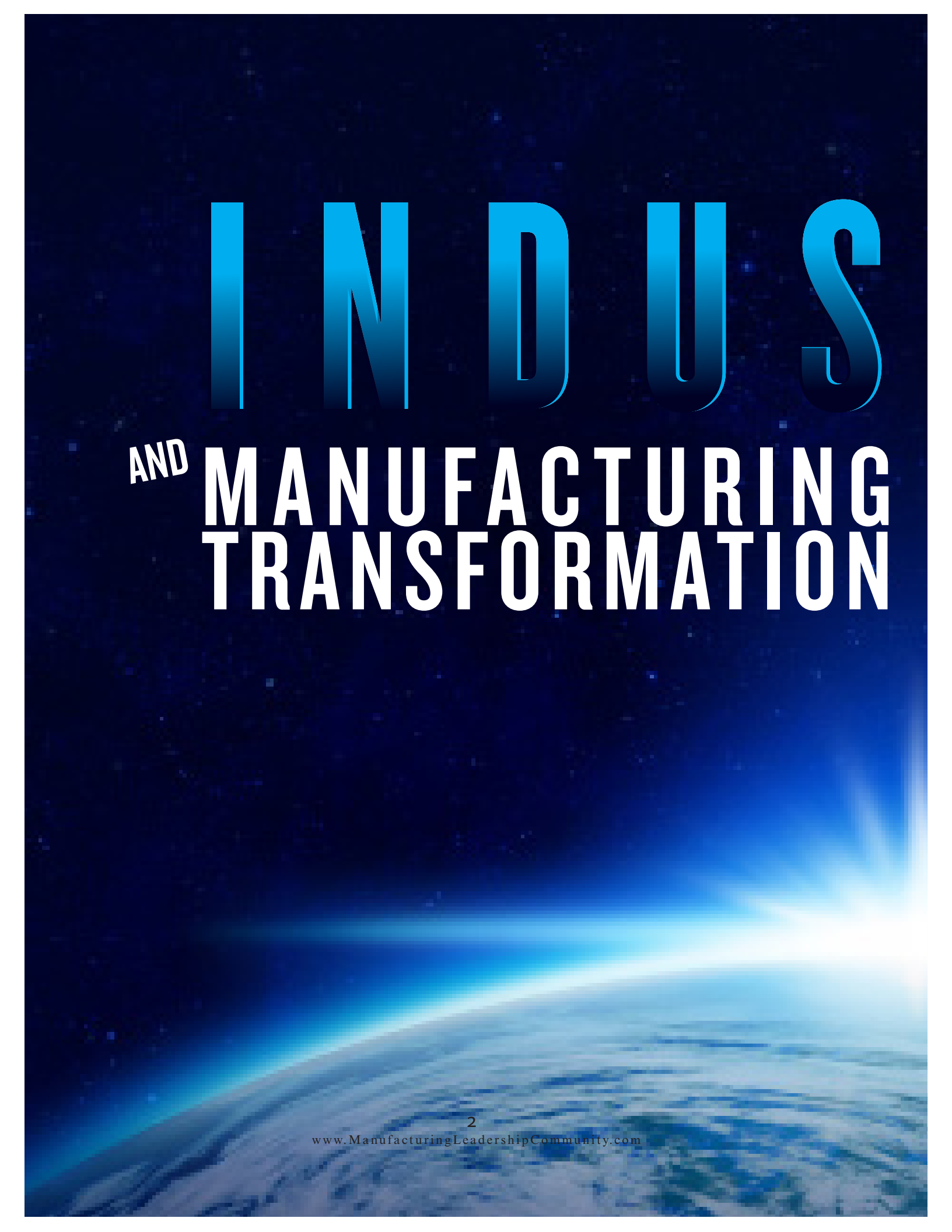




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TRANSFORMATION

TRY 4.0

Advances in cyber-physical systems promise to shatter the traditional operational paradigms and business models of today's manufacturing industry. The goal of new Industry 4.0 initiatives is to harness these new technologies to create smarter factories and new smart products in an effort to revitalize the manufacturing sector in the years ahead.

By Dr. Jay Lee, Behrad Bagheri,
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HE GLOBAL MANUFACTURING INDUSTRY HAS FACED CONTINUING challenges over recent years to improve manufacturing performance and sustainability. There are a number of reasons for this, including the entrenched traditional approaches of an increasingly aging workforce, the changing landscape of global manufacturing, and the relatively slow adoption of smart IT innovations in the production environment. In response, the U.S. government has now launched sweeping economic strategy plans that aim to strengthen the domestic manufacturing sector while also accelerating the growth of new product markets. As part of these plans, the U.S. government recently established a number of new manufacturing innovation institutes across the country, including innovation hubs that focus on Additive Manufacturing in Ohio, Low-Power Semiconductor Manufacturing in North Carolina, and Digital Manufacturing and Design Innovation (DMDI) and Light Weight Materials in Michigan. These dedicated new hubs followed the launch in 2013 of the Smart America Challenge, a collaborative initiative to showcase the use of advanced cyber-physical systems (CPS), also known as the Internet of Things (IoT), in different sectors, including manufacturing.

Across the Atlantic, Germany has also initiated a national effort that aims to trigger what it calls the fourth industrial revolution (or Industry 4.0). This is designed to help transform industrial production by the creation of "Smart Factories" and turn manufactured goods into "Smart Products" based on the extensive adoption of a cyber-physical systems platform. Industry 4.0, spurred by technological innovations in information technology, analytics, automation engineering, and other emerging technologies, promises to be the next, imminent industrial revolution for manufacturing.

Towards an Industry 4.0 Factory

Following its formal announcement at the 2013 Hannover Fair, there has been a huge amount of interest in Germany's Industry 4.0 initiative across the global manufacturing sector.

Industry 4.0 is a forward-looking plan developed under the auspices of the German Federal Government's High-Tech Strategy group. Focusing on cyber-physical systems, Industry 4.0 is regarded as the next-generation production framework for the fourth industrial revolution.

The German approach suggests that first revolution occurred by the end of the 18th century with the mechanization of manufacturing processes. Towards the start of the next century, in the second industrial revolution, electricity was utilized to power mass production of goods based on the division of labor (station-oriented). By the 1970s, the third industrial revolution emerged using electronics and information technology (IT) to achieve more automation of manufacturing operations. Industry 4.0 is the next step and its successful development and implementation promises to have game-

	Data Source	Today's Factory		Industry 4.0	
		Attributes	Technologies	Attributes	Technologies
Component	Sensor	Precision	Smart Sensors and Fault Detection	Self-Aware Self-Predict	Degradation Monitoring & Remaining Useful Life Prediction
Machine	Controller	Productibility & Performance	Condition-based Monitoring & Diagnostics	Self-Aware Self-Predict Self-Compare	Up Time with Predictive Health Monitoring
Production System	Networked System	Productivity & OEE	Lean Operations: Work and Waste Reduction	Self-Configure Self-Maintain Self-Organize	Worry-free Productivity

Table 1: Comparison of today's factory and an Industry 4.0 factory

changing business potential.

Industry 4.0 assumes that industrial production in the future will be characterized by a high degree of customization of products, supported by highly flexible and agile manufacturing operations.

Imagine having the capabilities that will allow a manufacturer to meet individual customer requirements while maintaining very low production volumes (as low as a single unit) and still make a profit. Manufacturing operations will be more forgiving to disruptions and last-minute changes to production. There will also be end-to-end transparency over the entire manufacturing process, allowing decision-makers to make informed decisions at any point in the production process.

In the longer term, the Industry 4.0 platform and its associated technologies also aim to address, and even solve, some important current global challenges, such as resource and energy efficiency, allowing

the entire value network to enjoy continuous resource productivity and efficiency gains, allowing workers to move away from low-value, routine tasks to work on more creative, value-added activities, and as a consequence, providing an opportunity for older workers to stay in their jobs longer while still being more productive. These long terms goals of Industry 4.0 clearly address many of the key challenges facing today's modern manufacturing industry.

Some of the main differences between the attributes of today's factories, versus Industry 4.0 factories, are illustrated in Table 1.

Smart Predictive Analytics

The potential repercussions of Industry 4.0 for the manufacturing sector are exciting and uplifting. With the right implementation, Industry 4.0 may well help revitalize manufacturing competitiveness.



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Dr. Edzel Lapira is president of Predictronics Corp., a spinoff of the NSF/IUCRC for Intelligent Maintenance Systems where he previously served as Associate Director at the University of Cincinnati. He's been involved in numerous predictive monitoring R&D projects with organizations including Caterpillar, Toyota, Nissan, TechSolve, and GlobalFoundries. He is also the co-inventor of several asset monitoring patents for wind turbines, industrial robots, and EV batteries.

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So, how can Industry 4.0 be implemented? The initial German initiative copiously mentioned the adoption of a cyber-physical system (CPS) platform in the implementation of this high-tech strategy. However, these recommendations provide limited insights about the definition and implementation guidelines for CPS. It is simply identified as a key priority for research, along with other foundational technologies of Industry 4.0.

Though CPS might still be in its infancy in Germany, the U.S. has already been investing in this approach in research activities, through the National Science Foundation, since 2006.

CPS is a complex engineering system that integrates physical, computation, and

networking/communication processes.

CPS can be illustrated as a physical device/object/equipment that is translated into cyberspace as a virtual model. With networking capabilities, the virtual model can monitor and control its physical aspects, while the physical objects send data to update and improve the virtual model. In order to build and maintain such virtual models, intelligent predictive analytics are utilized. Such advanced analytics can then be used to intelligently process all the data that is generated from all IoT and networked sensors (IoS). If CPS is the foundation for a successful implementation of Industry 4.0, then so are the advanced predictive analytics aspects of CPS.

One area that is already leveraging such intelligent algorithms is in prognostics and health management (PHM). This involves estimating the condition of a manufacturing asset/process, the identification of in-

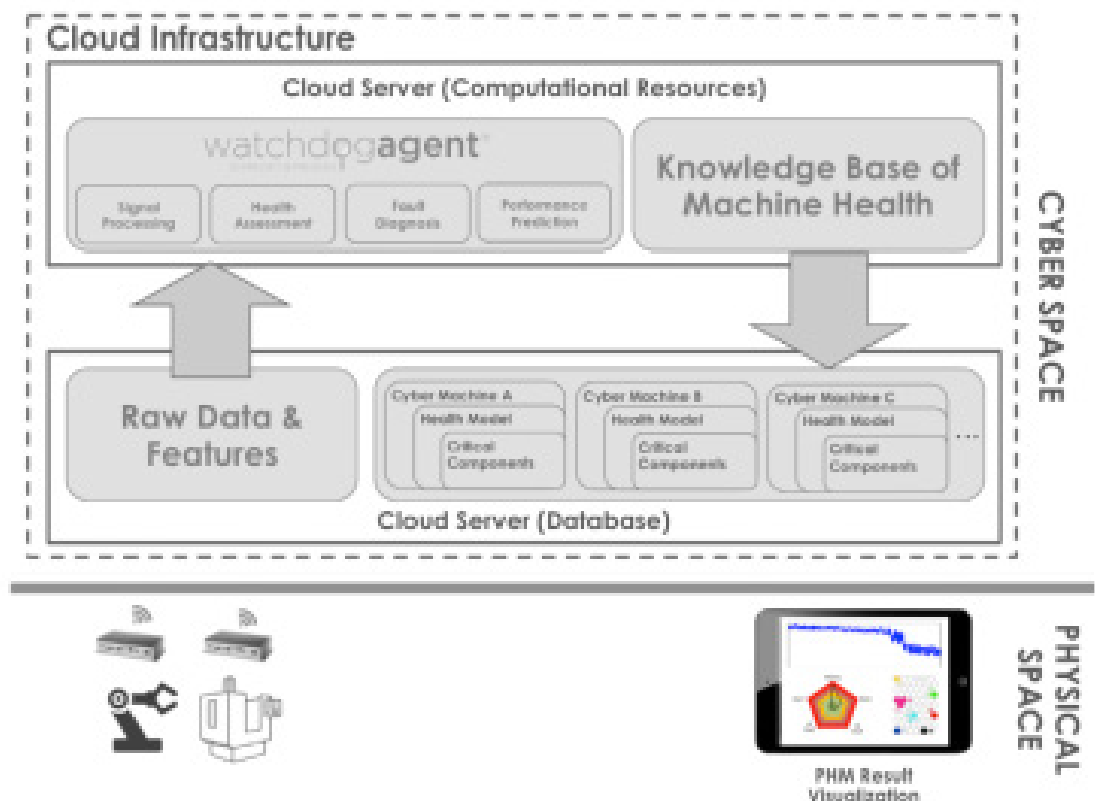


Figure 1. Cyber-physical modeling for manufacturing asset monitoring

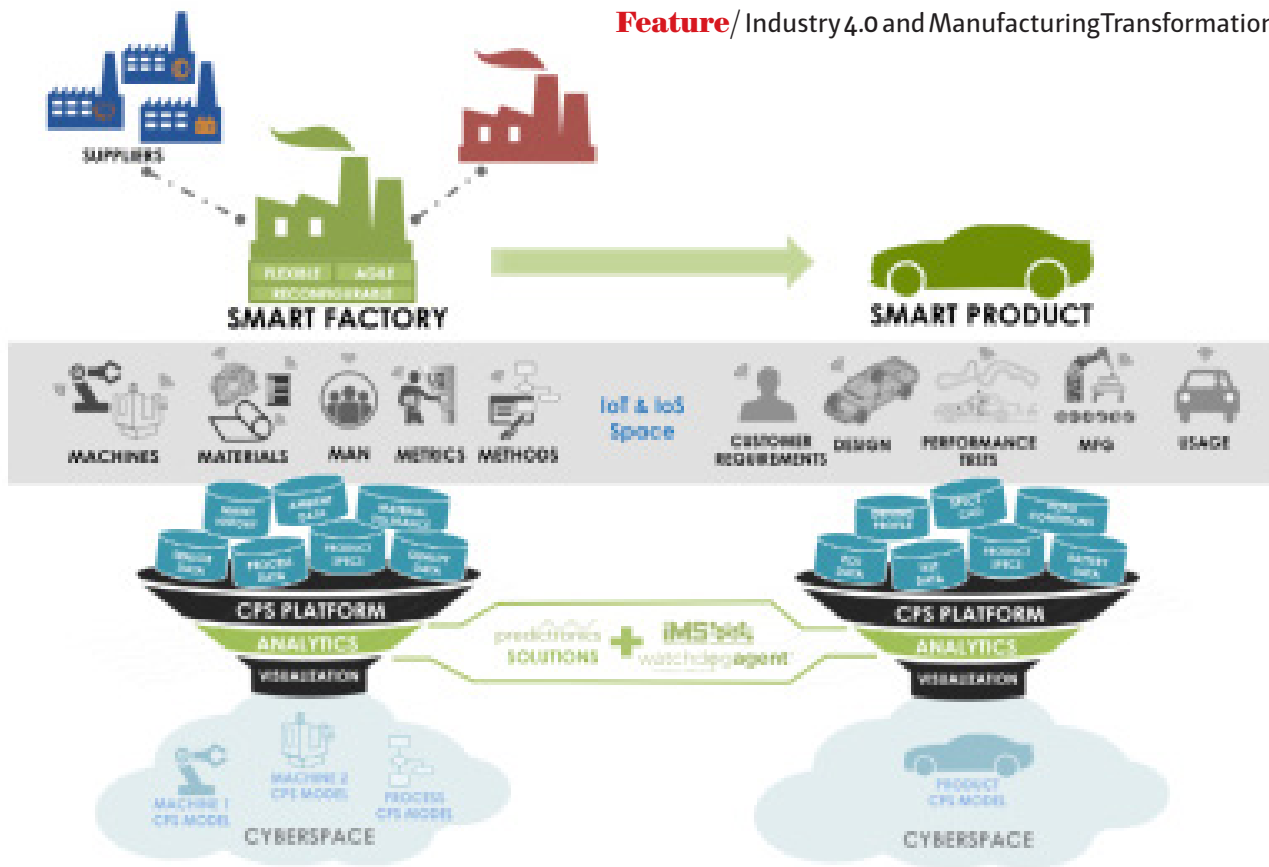


Figure 2. Framework for using cyber-integrated analytics in the Industry 4.0 platform

ipient or early signs of degradation, the inference of the next failure event, and the diagnosis of the impending fault event so that appropriate proactive maintenance action can be performed to avoid potentially catastrophic and expensive machine downtime altogether (see Figure 1).

Cyber-integrated Predictive Analytics for Industry 4.0

The goals of industry 4.0 are to improve the overall competitiveness of the manufacturing sector while accelerating the growth of new areas of business. As a technology investment, Industry 4.0 therefore promises to deliver a twofold benefit. An Industry 4.0 platform may not only provide improvements in how companies operate their manufacturing assets by creating so-called “Smart Factories,” it also opens the doors to new markets through the development of inno-

vative smart products and even services. In order to attain the benefits and goals of Industry 4.0, essential platform components need to be in place and harmoniously utilized, as illustrated in Figure 2.

The Smart Factory should be agile, flexible and reconfigurable, and easily adaptive to changes in market demands and customer requirements. It is also expected to work collaboratively with other factories and even suppliers. Manufacturing transparency is therefore key to seamlessly exchanging information and resources across, and between, Smart Factory networks.

An Industry 4.0 Example

Predictronics Corporation is a spinoff from the National Science Foundation (NSF) Industry-University Cooperative Research Center for Intelligent Maintenance



Further Reading:

1 Lee, J., Kao, HA, Industry 4.0 in Big Data Environment, *German Harting Magazine* 26, 2013, pp.8-10.

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4 Lee J, Lapira E. Predictive Factories: The Next Transformation. *Manufacturing Leadership Journal*. Feb 2013;

5 Lee J, Lapira E, Yang S, Kao HA. Predictive Manufacturing System Trends of Next Generation Production Systems. *Proceedings of the 11th IFAC Workshop on Intelligent Manufacturing Systems*. 2013;150-156.

6 Lee, J, Bagheri B, and Kao H. A Cyber Physical Systems Architecture for Industry 4.0-based Manufacturing Systems. *SME journal of Manufacturing Letters* (2014).

Systems (IMS) at the University of Cincinnati. Since 2001, the IMS Center has been developing a toolbox of predictive analytics, called the Watchdog Agent®, and validating the approach with consortium members from different market segments. Predictronics has licensed this technology to build predictive monitoring solutions such as the Factory Sentinel (see Figure 3).

The first version of Factory Sentinel includes a health-monitoring solution for industrial robots. It will provide an early detection of incipient failure through degradation assessment, instead of the traditional fault-detection solutions that sound an alarm when the fault has already occurred. This will enable manufacturers to proactively perform maintenance before the actual fault occurs without disrupting expected production schedules.

The software solely uses data collected from the robot controller; therefore no additional sensors need to be installed. Signature information from the torque signal from each axis on the robot creates a fingerprint of the robot's behavior. Through advanced analysis methods and a comparison with the baseline behavior, a single degradation health index can be calculated for each robot, providing easy-to-read information for operators and maintenance technicians. Contribution plots can also be used to isolate a particular problem on a given robot axis. This can help aid in repair and troubleshooting for the maintenance team when some early signs of robot degradation are found.

Examples of the robot health monitoring software are highlighted in Figure 4 and Figure 5, which show the health value, contribution plots, and other visual information. With greater transparency of the robot's health condition, there should be fewer surprises and less unplanned downtime for a factory and its robot fleet.

The first versions of Factory Sentinel are currently being tested by early adopters in the automotive manufacturing and assembly industries. This will enable the developers to test out the system on robots from different OEMs, as well as get customer feedback on software features

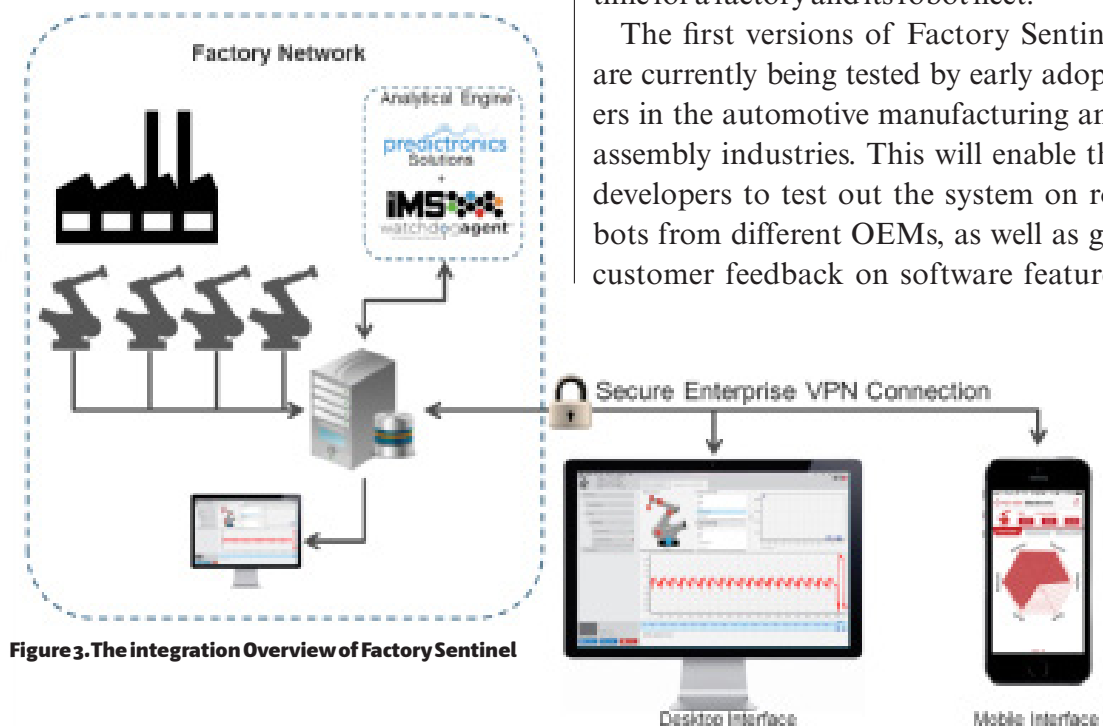


Figure 3. The integration Overview of Factory Sentinel

and functionalities. Future functionalities include integration with Enterprise Resource Planning (ERP) and Manufacturing Execution Systems (MES), mobile applications for visualizing the health results, and the inclusion of automated trend and prediction tools into the software.

The Industry 4.0 Journey

Technological innovations in products and processes have been catalytic to the transformation of manufacturing for several centuries. The astounding advances in information technology, automation engineering, cloud infrastructure, analytics, new manufacturing technologies such as additive manufacturing, and even social media, have gradually started to reshape how manufacturing businesses are now being conducted, shattering traditional business models and paradigms.

In Industry 4.0, traditional production facilities are converted into Smart Factories that in turn make Smart Products. But the journey towards Industry 4.0 is expected to be gradual and evolutionary. Many of the basic underlying technologies still need to be researched. Existing manufacturing infrastructures will have to be adapted to the requirements of Industry 4.0. New tools, technologies and methodologies to create effective CPS

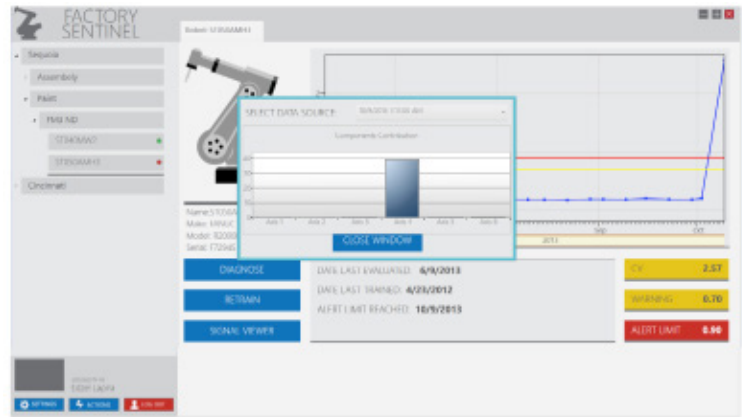


Figure 4. Health Value Plot and Contribution Values for Factory Sentinel Software

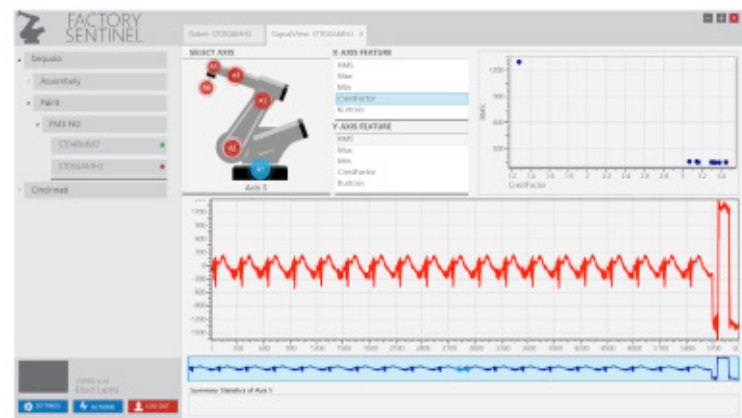


Figure 5. Signal and Feature View Interface - Factory Sentinel Software

models will need to be developed.

But if adopted and implemented successfully, Industry 4.0, using cyber-integrated advanced analytics, will help companies improve their global competitiveness, restore their domestic manufacturing industry, and break ground in new market opportunities. **M**

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